

Technical Report: Anatomical Adaptations of Cheloniodea

Sea turtles (**superfamily Cheloniodea**) possess highly specialized anatomical features evolved for a strictly marine existence. The most prominent structure is the **carapace**, a streamlined shell fused to the vertebrae and ribs. Unlike terrestrial tortoises, sea turtles cannot retract their heads into their shells; this trade-off allows for a more hydrodynamic profile, reducing drag during long-distance migration.

The skeletal system is complemented by modified appendages. The forelimbs are elongated into powerful, wing-like **flippers** used for propulsion, while the hind flippers act as rudders for stabilization and steering. Internally, sea turtles exhibit significant physiological adaptations for deep diving. Their lungs are capable of rapid gas exchange, and they possess high levels of myoglobin in their muscle tissue to store oxygen.

A critical cranial adaptation is the presence of **salt glands** located behind the eyes. These glands excrete excess salt ingested from seawater, maintaining osmotic balance. Furthermore, the esophagus of many species, such as the Leatherback, is lined with sharp, keratinized **papillae**. These structures point inward to trap prey—specifically gelatinous zooplankton—while allowing excess water to be expelled before ingestion. This suite of morphological traits ensures survival in diverse oceanic thermoclines.